

Hannah T. Nguyen

htn7254@utulsa.edu

University of Tulsa
800 South Tucker Drive
Tulsa, OK 74104

Cyber Corps Program
Department of Electrical and
Computer Engineering
Tandy School of Computer
Science

EDUCATION

University of Tulsa (NSA Center of Excellence)

M.S. Computer Engineering (Cyber Operations Specialization) GPA: 4.000

B.S. Electrical and Computer Engineering (Cyber Operations Specialization) GPA: 4.000

Tulsa, Oklahoma

May 2028 (expected)

May 2027 (expected)

Union High School

High School Diploma GPA: 4.96 (Weighted)

Tulsa, Oklahoma

May 2023

COMPUTER ENGINEERING/SCIENCE AND MATHEMATICS COURSEWORK

Spring 2025: Electronics II and Laboratory, Classical Control Theory and Applications, Operating Systems

Fall 2025: Electric Power Systems, Signal & Linear Systems, Electronics I, PCB Design Principles and Practice, Embedded Systems in C

Completed Courses: Hardware Reverse Engineering, Computer Organization and Microprocessors, Introduction to Electromagnetics, Linear Algebra and Matrix Theory, Data Structures, Electrical Circuit Analysis, Basic Electrical Measure Laboratory, Differential Equations, Digital Systems Principles, Discrete Mathematics, Fundamentals of Algorithms and Computer Applications, General Physics I-III, Statistical Methods for Scientists and Engineers, Radio Frequency Signal Reverse Engineering, Introduction to Programming and Problem Solving, Ethics in Computer Science and Engineering, Calculus I-III, Introduction to Statistics

LANGUAGES AND PLATFORMS

Operating Systems: Windows 10/11, Linux, Ubuntu

Programming Languages: Java, Python, CSS, HTML, JavaScript, Lua, RobotC, Latex, MATLAB, MIPS, VHDL, Vue3, Vuetify3

Software: Autodesk Inventor, Autodesk Fusion 360, AutoCAD, Microsoft Office, KiCAD, Hercules, MATLAB Simulink, Simscape, Visual Studio Code, Arduino IDE, Soldering, Altium Designer

Spoken Languages: English, Vietnamese

EMPLOYMENT

University School, University of Tulsa, Tulsa, Oklahoma

Fall 2025 – Present

Lab Assistant: Assisted with 7th grade Computer Science and 8th grade Computer Engineering classes, providing classroom support and guidance. Additionally, supported U-School's STEM Club by helping organize activities and engaging students in hands-on projects.

U.S. Government, Washington D.C.

Summer 2025

Software Developer Intern: Developed and designed an application to assist with data triage of a collection with the utilization of AI. Specialized in engineering, development, integration, and product evaluation services for the workgroup platform. Provided critical support for the infrastructure to allow for communications and collaboration.

NCFI Laboratory, University of Tulsa, Tulsa, Oklahoma

Fall 2023 – May 2025

Undergraduate Researcher: Conducted research on the development and applications of alternative reality software and tools in order to quickly and affordably re-create digital versions of crime scenes as well as reconstruction large areas to allow for digital analysis of protection locations during event planning.

TURC Program, University of Tulsa, Tulsa, Oklahoma **Summer 2024**

Volunteer Java Programming Instructor: Co-taught two Java programming classes to 37 incoming freshmen. The courses gave the students a head start in their computer science education, enabling them to bypass the first two classes, Introduction to Programming and Problem Solving and Fundamentals of Algorithms and Computer Applications, and take advanced courses during their first semester in college.

Junior TURC Program, University of Tulsa, Tulsa, Oklahoma **Summer 2022**

High School Researcher: Helped update the Rayzor Hall keycard system from an Arduino implementation to an STMicroelectronics implementation. Also helped design a physical adaptation of the VR game “Keep Talking and Nobody Explodes.”

Information Technology Department, Union Public Schools, Tulsa, Oklahoma **Fall 2022 – Spring 2023**

Information Technology Intern: Provided in-person assistance to students with information technology problems. Repaired hardware and installed software.

CERTIFICATIONS

SANS GIAC Incident Handler Certification (GCIH) (January 2025 – January 2029): The GIAC Incident Handler (GCIH) certification validates a practitioner's ability to detect, respond, and resolve computer security incidents using a wide range of essential security skills. GCIH certification holders have the knowledge needed to manage security incidents by understanding common attack techniques, vectors and tools, as well as defend against and respond to such attacks when they occur.

TS/SCI Clearance

PROJECTS

Personal Portfolio Website: Project involved designing and deploying a personal portfolio website using HTML, CSS, JavaScript, and Bootstrap. Implemented responsive layouts, interactive features, and version control with GitHub Pages to showcase projects and technical skills. (TheNguyen07.github.io)

Projectile Trajectory Simulation (Fall 2025): Project involved programming in MARS MIPS assembly and C to calculate projectile time-of-flight, maximum height, and trajectory angle based on user input. Applied physics equations of motion to model artillery firing scenarios, validated results across both implementations, and documented methodology, findings, and error analysis in a professional report.

Reverse Engineer PCB (Fall 2025): Reverse engineered a custom PCB and implemented the functionality on an FPGA board using VHDL.

Single-Port Read/Write Memory Design (Spring 2024): Lab focused on designing and constructing a 16×4-bit memory device using a 64-bit shift register, address counter, and supporting logic. Implemented address comparison with XNOR/AND gates, clock division using T flip-flops, and read/write functionality through multiplexers and D flip-flops. Verified functionality with LEDs and 7-segment displays, and debugged timing and hardware issues to ensure reliable operation.

University RFID Analysis (Fall 2023): Conducted a capstone research project on reverse engineering TU ID card systems, identifying three software and hardware vulnerabilities. Proposed and documented security solutions including password protection with lock bits, reinforced hardware design, and card number obfuscation.

AWARDS AND ACTIVITIES

IEEE TU Chapter (May 2024 – Present): President. The Institute of Electrical and Electronics Engineers gives students a community of peers, and a connection to faculty and industry professionals who drive innovation in countless technical fields. Worked with the Department Chair to help improve the college.

Student Government Association (September 2024 – January 2025): College of Engineering and Computer Science Senator.

Eta Kappa Nu (HKN) (April 2024– Present): Honor society member.

University of Tulsa President's Honor Roll (January 2024 - Present): Given to full-time students who maintain a term GPA of 4.0.

Alan Paller Cyber Scholarship (January 2024 – Present): Prestigious scholarship for excellence in cyber studies and research, and a commitment to serve with the federal government or OPM-approved organizations. The scholarship provides a stipend and the opportunity to take SANS courses and certification exams at no cost.

University of Tulsa Presidential Scholarship (August 2023 – Present): Full tuition scholarship awarded to high-achieving students who have demonstrated academic excellence in high school.

AP Scholar with Distinction (July 2023): Honor accorded to students who receive an average score of at least 3.5 on all AP Exams taken, and scores of 3 or higher on five or more of the exams.

Oklahoma Academic Scholar (May 2023): Honor accorded to students who accumulate a minimum grade point average of 3.7 or were in the top 10% of their graduating class and obtained a minimum of 27 ACT composite score or a 1220 SAT score.

Tulsa Engineering Scholarship (May 2023 – Present): Scholarship awarded to high-achieving students who major in an engineering field.

FTC State Championship Control Award Sponsored by ARM, Inc. (February 2023): Awarded to high school teams that use sensors and software to increase robot functionality. Teams must demonstrate innovative thinking to solve challenges such as autonomous operation and improving mechanical systems using sensors and intelligent control schemes.

Salutatorian (May 2023): Honor accorded to students in the Top 2% of the 2023 Union High School graduating class of 1,075 students.

REFERENCES

Dr. Sujeet Shenoi, F.P. Walter Professor of Computer Science and Professor of Chemical Engineering, University of Tulsa, 800 South Tucker Drive, Tulsa, OK 74104; tel: (918) 344-1288; email: sujeet@utulsa.edu

Assistant to the Special Agent in Charge Steven Baskerville, Supervisor, U.S. Secret Service National Computer Forensics Institute (NCFI) Laboratory, University of Tulsa, 800 South Tucker Drive, Tulsa OK 74104; tel: (202) 805-0459; email: sbaskerville@usss.dhs.gov

Technical Special Agent Timothy Hunt, R&D, University Projects and Accreditation Section, U.S. Secret Service National Computer Forensics Institute (NCFI) Laboratory, University of Tulsa, 800 South Tucker Drive, Tulsa, OK 74104; tel: 202-507-3698; email: timothy.hunt@usss.dhs.gov

Dr. Nathan Hutchins, Applied Assistant Professor of Electrical and Computer Engineering, University of Tulsa, 800 South Tucker Drive, Tulsa OK 74104; email: nathan-hutchins@utulsa.edu

Dr. Ian Riley, Instructor of Computer Science, University of Tulsa, 800 South Tucker Drive, Tulsa, OK 74104; email: isr413@utulsa.edu

Dr. John Hale, Tandy Professor of Bioinformatics and Computational Biology and Chair of the Tandy School of Computer Science, University of Tulsa, 800 South Tucker Drive, Tulsa, Oklahoma 74104; tel: (918) 631-2745; email: john-hale@utulsa.edu

Professor Douglas Jussaume, Adjunct Professor of Electrical and Computer Engineering, University of Tulsa, 800 South Tucker Drive, Tulsa OK 74104; email: douglas-jussaume@utulsa.edu